Reading list

Computational macrocyclic peptide design

A. Foundations — macrocyclic peptides, ncAAs and genetic code reprogramming

Katoh, T. & Suga, H. (2022). In Vitro Genetic Code Reprogramming for the Expansion of Usable Noncanonical Amino Acids. Annual Review of Biochemistry 91, 221–243.
https://doi.org/10.1146/annurev-biochem-040320-103817

> **Relevance.** Covers flexizyme-mediated tRNA acylation, the FIT system, and the practical limits on which non-canonical building blocks the ribosome will accept — including D-, N-methyl-, β- and γ-amino acids, and the engineering of ribosomes, EF-Tu and tRNAs needed for multiple and consecutive incorporations.

Huang, Y., Wiedmann, M. M. & Suga, H. (2019). RNA Display Methods for the Discovery of Bioactive Macrocycles. Chemical Reviews 119(17), 10360–10391.
https://pubmed.ncbi.nlm.nih.gov/30395448/

> **Relevance.** Comprehensive background on mRNA display, RaPID and flexizyme technology.

Muttenthaler, M., King, G. F., Adams, D. J. & Alewood, P. F. (2021). Trends in peptide drug discovery. Nature Reviews Drug Discovery 20, 309–325.
https://www.nature.com/articles/s41573-020-00135-8

> **Relevance.** Commercial and clinical framing: where peptide therapeutics sit against small molecules and biologics, the recurring liabilities (proteolytic stability, permeability, half-life, route of administration), and how chemical modification addresses them.

B. mRNA display datasets and experimental mutational scanning

Discovery of Macrocyclic Peptide Binders, Covalent Modifiers, and Degraders of a Structured RNA by mRNA Display (2025). PMID 40951976 / PMCID PMC12464979.
https://pmc.ncbi.nlm.nih.gov/articles/PMC12464979/

Rogers, J. M., Passioura, T. & Suga, H. (2018). Nonproteinogenic deep mutational scanning of linear and cyclic peptides. PNAS 115(43), 10959–10964.
https://www.pnas.org/doi/epdf/10.1073/pnas.1809901115

C. Peptide representation learning and language models

Feller, A. L. & Wilke, C. O. (2025). Peptide-Aware Chemical Language Model Successfully Predicts Membrane Diffusion of Cyclic Peptides (PeptideCLM). Journal of Chemical Information and Modeling 65(2), 571–579.
https://pubs.acs.org/doi/abs/10.1021/acs.jcim.4c01441

Wang, L. et al. (2024). PepDoRA: A Unified Peptide Language Model via Weight-Decomposed Low-Rank Adaptation. arXiv:2410.20667.
https://arxiv.org/pdf/2410.20667

Systematic benchmarking of 13 AI methods for predicting cyclic peptide membrane permeability (2025). PMCID PMC12392647.
https://pmc.ncbi.nlm.nih.gov/articles/PMC12392647/

Bhat, S. et al. (2025). De novo design of peptide binders to conformationally diverse targets with contrastive language modeling (PepPrCLIP). Science Advances. PMCID PMC11753435.
https://pmc.ncbi.nlm.nih.gov/articles/PMC11753435/

Chen, L. T. et al. (2025). Target sequence-conditioned design of peptide binders using masked language modeling (PepMLM). Nature Biotechnology 44, 1002–1010.
https://pmc.ncbi.nlm.nih.gov/articles/PMC13271882/

D. Structure-based de novo design of peptides and binders

Huang, B. et al. (2024). De novo design of miniprotein antagonists of cytokine storm inducers. Nature Communications 15, 7064.
https://pmc.ncbi.nlm.nih.gov/articles/PMC11329760/

Rettie, S. A. et al. (2025). Accurate de novo design of high-affinity protein-binding macrocycles using deep learning (RFpeptides). Nature Chemical Biology 21(12), 1948–1956.
https://www.nature.com/articles/s41589-025-01929-w

Latent Labs Team (2025). Latent-X: An Atom-level Frontier Model for De Novo Protein Binder Design. arXiv:2507.19375.
https://arxiv.org/pdf/2507.19375

> **Relevance.** Co-generates all-atom structure and sequence for binder and target together, reporting macrocycle hit rates above 90% on its benchmark targets with only 30–100 designs tested.

BoltzGen: Toward Universal Binder Design (2025). bioRxiv 2025.11.20.689494.
https://www.biorxiv.org/content/10.1101/2025.11.20.689494v2.full.pdf

> **Relevance.** A generalist binder design model spanning multiple modalities and target classes.

E. Physics-based modelling of non-canonical residues (Rosetta)

Barlow, K. A. et al. (2018). Flex ddG: Rosetta Ensemble-Based Estimation of Changes in Protein–Protein Binding Affinity Upon Mutation. Journal of Physical Chemistry B. PMCID PMC5980710.

https://pmc.ncbi.nlm.nih.gov/articles/PMC5980710/

Relevance. An ensemble protocol for estimating the binding free energy change on mutation, substantially more reliable than single-structure ΔΔG for small interfaces.

FakeRotLib: Expedient Noncanonical Amino Acid Parametrization in Rosetta. Journal of Chemical Information and Modeling 65(16), 8397.

https://pubs.acs.org/jcisd8/article/65/16/8397/3687663/FakeRotLib-Expedient-Noncanonical-Amino-Acid

Relevance. A fast route to generating usable rotamer libraries for new ncAAs to address the main bottleneck in bringing a novel building block into Rosetta.

Renfrew, P. D., Choi, E. J., Bonneau, R. & Kuhlman, B. (2012). Incorporation of noncanonical amino acids into Rosetta and use in computational protein–peptide interface design (MakeRotLib). PLoS ONE 7(3), e32637.

https://journals.plos.org/plosone/article/file?id=10.1371/journal.pone.0032637&type=printable

Relevance. The original method paper for ncAA parameterisation in Rosetta. Read alongside FakeRotLib to understand what the faster method approximates.

RosettaCommons documentation: Using NCAAs in Protein–Peptide Interface Design.

https://docs.rosettacommons.org/demos/latest/public/using_ncaas_protein_peptide_interface_design/README

Relevance. Practical, hands-on protocol.

Meiler Lab. Non-canonical peptide and macrocycle design (slide deck, 2025).

https://meilerlab.org/wp-content/uploads/2025/11/peptide_design_slides.pdf

Relevance. A current overview of the macrocycle and ncAA design toolchain from one of the main groups working on it.

Dunbrack, R. L. Jr. (2002). Rotamer libraries in the 21st century. Current Opinion in Structural Biology 12, 431–440.

https://dunbrack.fccc.edu/bbdep2010/papers/Dunbrack_CurrOpin_Rotamers_2002.pdf

F. Methods and general machine learning

Joint Embedding Predictive Architecture for self-supervised pretraining on polymer molecular graphs (2025). arXiv:2506.18194.

https://arxiv.org/pdf/2506.18194

Lakshminarayanan, B., Pritzel, A. & Blundell, C. (2017). Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. arXiv:1612.01474.

https://arxiv.org/pdf/1612.01474

Gonzalez-Duque, M. Bayesian Optimization using Gaussian Processes: an introduction (blog post).
https://www.miguelgondu.com/blogposts/2023-07-31/intro-to-bo/

Gonzalez-Duque, M. Do Gaussian Processes scale well with dimension? (blog post).
https://www.miguelgondu.com/blogposts/2024-03-16/when-does-vanilla-gpr-fail/